Flood Risk Prediction System

Civil Engineering Machine Learning Project — 2025-2026

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Data Sources:

We used two real public government datasets. USGS (United States Geological Survey) provided daily river discharge and stage measurements for 11 gauge stations across the US from 2018 to 2024, giving us around 28,000 daily observations. NOAA (National Oceanic and Atmospheric Administration) provided daily weather data including precipitation, temperature, wind speed, and snowfall, matched to each river station.

Approach:

Given the last 7 days of river discharge and weather at a gauge station, we predict the flood risk level for the next day: Normal (below p75), Medium (between p75 and p90), or High (above p90 — flood risk). Thresholds are computed per station from historical data so each river has its own calibrated flood levels.

Methods:

We engineered 38 features including discharge lag values, rolling statistics, normalized discharge relative to each river's historical levels, cyclic seasonal encoding, precipitation rolling sums, and weather variables. We trained two models on a strict chronological split (70% train / 15% validation / 15% test) with no random shuffling to prevent data leakage. The baseline is a Logistic Regression pipeline with StandardScaler. The strong model is LightGBM with 300 gradient boosted trees and balanced class weights. Both models use probability calibration via isotonic regression on the validation set. Predictions are served through a FastAPI backend with a Next.js dashboard.

Results:

Model	Test F1 Score
Logistic Regression (baseline)	0.867
LightGBM (strong model)	0.904

LightGBM significantly outperforms the baseline and provides reliable next-day flood warnings across all 11 river stations.

How to Run:

```bash
pip install -r requirements.txt
python run_api.py        # Terminal 1 — starts API at localhost:8000
cd frontend1 && npm install && npm run dev     # Terminal 2 — dashboard at localhost:3000
```